

CS3 Case Study Rubric - Sentiment Analysis of U.S. State of the Union Addresses

Project Topic: NLP Sentiment Analysis of Speeches

Audience: 2nd-year UVA Data Science students

Project Source: Group Project DS4002 - Corner Data Co.

Why am I doing this? The purpose of this case study is to guide you through building and evaluating a sentiment analysis pipeline that examines whether U.S. presidents speak differently during times of national crisis. You will work with a real historical dataset of State of the Union addresses spanning 1850–2025. Using the Hugging Face Transformers library, you will apply a pretrained NLP model to classify sentiment at the sentence level, aggregate results at the speech level, and use statistical testing to compare crisis vs. non-crisis speeches. The goal is to help you practice data cleaning, sentiment analysis, and statistical evaluation while engaging with a meaningful real-world question about presidential rhetoric.

What am I going to do?

- Final product: Recreate GitHub repository results
- Run exploratory data analysis on the dataset
- Run sentence-level sentiment analysis using the Hugging Face Transformers library
- Aggregate sentiment results at the speech level
- Compare crisis vs. non-crisis sentiment using a two-sample t-test
- Visualize results and interpret findings

Rubric Metrics: meets expectations when

Formatting & Repository Structure	● GitHub repository is clean, well-organized, and easy to navigate. ● Student clones or downloads the repository and verifies that all required files are present ● Student reads the README before running any scripts to understand the project structure and reproduction steps. ● Student verifies that the following items are present: ○ README.md ○ Hook document (PDF) ○ Rubric (PDF) ○ DATA folder containing SOTU_data_final.csv ○ SCRIPTS folder containing project_1_edu.py, sentiment_model.py, and stat_analysis.py, ○ OUTPUTS folder, and Supplemental Materials folder with two sources.
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Data Preparation & Exploration	• Student loads SOTU_data_final.csv successfully. Student runs project_1_edu.py from the SCRIPTS folder. • The following plots display on screen: ◦ crisis vs. non-crisis speech counts ◦ average word count by crisis flag ◦ crisis years by type (military vs. economic) ◦ frequency of crises mentioned in SOTU addresses. • Student notes at least two observations about the data before moving on. • Student verifies the dataset contains 231 rows spanning 1850–2025.
Crisis Labeling & Feature Engineering	• Student reviews the is_crisis and crisis_name columns in SOTU_data_final.csv to understand how crisis years were defined. • Student can explain the difference between military and economic crises as classified in the dataset. • Student identifies at least one challenge involved in defining a crisis year, for example, crises do not always have clear start and end boundaries. ◦ Optionally, if the student disagrees with any classification, they may update the is_crisis or crisis_name columns and rerun the analysis to explore how definitional choices affect results.
Sentiment Analysis & Modeling	• Student runs sentiment_model.py from the SCRIPTS folder. • The model is applied at the sentence level to all speeches. ◦ Each speech is classified as POSITIVE if more than 50% of its sentences are positive, or NEGATIVE if 50% or fewer are positive. • Student verifies that sentiment_analysis_results.csv has been created in the OUTPUTS folder before proceeding to the next step.
Statistical Evaluation & Visualization	• Student runs stat_analysis.py from the SCRIPTS folder. Student verifies that all six visualizations and analysis_summary.txt have been generated in the OUTPUTS folder. • Student reports the t-statistic and p-value from the two-sample t-test and interprets whether the result is statistically significant. • Student writes a short reflection addressing the research question, " Does crisis change the sentiment of State of the Union addresses?, and proposes at least one next step for future research.